How to Kernelize SVM

$$\mathcal{L}_D(\alpha) = \sum_{i=1} \alpha_i - \frac{1}{2} \sum_{j=1} \sum_{i=1} \alpha_i \alpha_j y_j y_i x_j^T x_i$$

$$w = \sum_{i=1} \alpha_i y_i x_i$$

$$b = -\frac{1}{2} \left[\max_{i:y_i=-1} w^T x_i + \min_{i:y_i=+1} w^T x_i \right]$$

Class of vector **Z** ⇒ *sign*($W^T Z + b$)

Remember :

Mapping Instead of using X_i (of dimension ***d***) use $\varphi(X_i)$ *(with higher dimension)*

$$\varphi(x_i)^T \varphi(x_j) = \varphi(x_i)^T \varphi(x_j)$$

Kernelization Instead of using $\varphi(X_i)$, $\varphi(X_j)$ use $\mathbf{K(X_i^T, X_j^T)}$ *(Where K is one of kernel functions)*

$$\mathcal{L}_D(\alpha) = \sum_{i=1} \alpha_i - \frac{1}{2} \sum_{j=1} \sum_{i=1} \alpha_i \alpha_j y_j y_i x_i^T x_j$$

$$w = \sum_{i=1} \alpha_i y_i x_i$$

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